developed by Borland, was once a very popular IDE for Pascal.

Now that we have learned a little bit about the history of Pascal, here comes the important question: How can we execute Pascal or Object Pascal programs in Linux? There was a proposal to develop a compiler called GNU Pascal, which was supposed to be a part of GCC (GNU Compiler Collections). But the project is long delayed and doesn't seem to become a reality anytime soon. But there's no need to worry. There is a compiler called Free Pascal, which can be used to compile both Pascal and Object Pascal programs. Free Pascal compiler is free and open source software licensed under the GNU General Public License. The compiler can be installed with the command:

```
sudo apt install fp-compiler
```

…in Ubuntu. I am sure the installation will only require one or two commands in other Linux distributions also. Now that we have installed the compiler in our system, let us use a small Pascal program to test the Free Pascal compiler. Consider the program called *message.pp* shown below. Notice that, in addition to the extension *.pp*, Pascal/Object Pascal programs can also have extensions like *.p*, *.pas*, etc.

```
{ One Commenting Style in Pascal }
(* A Second Commenting Style in Pascal
*)

Program HWP(output);
begin
  writeln('Hello World from Pascal');
end.
```

The program can be compiled with the command *fpc message.pp*. This command will generate an executable file called *message*. This file can be executed with the command */message*. Figure 2 shows the compilation and execution of the program *message.pp*. Notice that the program *message.pp*

```
File  Edit  View  Search  Terminal  Help
# fpc message.pp
Free Pascal Compiler version 3.2.0 [2020/07/07] for x86_64
Copyright (c) 1993-2020 by Florian Klaempfl and others
Target OS: Linux for x86-64
Compiling message.pp
Linking message
8 lines compiled, 0.1 sec
# ./message
Hello World from Pascal
#
```

Figure 2: Output of the program *message.pp*

is a minor modification to the classical program to print the 'Hello World' message on the screen. The program also shows the two different commenting styles available in Pascal.

But, what makes Pascal unique among classical programming languages is the availability of a powerful IDE called Lazarus. Of course, the popular IDE called Code::Blocks can be used to work with Fortran, and an IDE called OpenCobolIDE can be used to work with COBOL. But Lazarus is a dedicated Pascal IDE and very powerful, thereby making it a true enhancement to software development with Pascal. Lazarus IDE is free and open source software licensed under the GNU General Public License. It uses the Free Pascal compiler in the background. Figure 3 shows the logo of

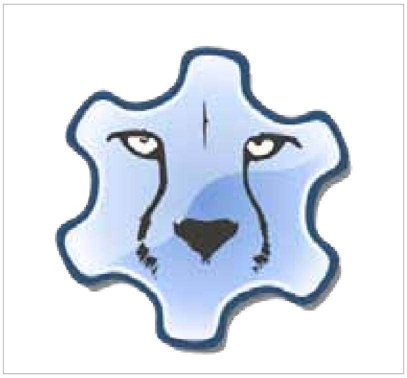

Figure 3: The logo of Lazarus IDE

Lazarus IDE. Remember that the Delphi dialect used in the Delphi IDE is the most prevalent dialect of Object Pascal (and Pascal for that matter) nowadays. Here comes an important and useful feature of the Lazarus IDE. There is a conversion tool which converts the Linux dialect of Object Pascal to the Delphi dialect, in Lazarus IDE. Thus any program written and tested in the Linux platform can be easily ported to the Microsoft Windows platform also.

Here's where we end this article and the series on classical programming languages. In this series, we saw why three programming languages — Fortran, COBOL, and Pascal — each more than half a decade old, are still relevant. But before we sign off, what predictions can we make about the future of these three programming languages? I am sure that these languages are not going to die out in the near future. Of course, like what the quote 'Everything that has a beginning has an end' says, eventually at some point in time these languages may also die out. When this happens, a major reason would be the unavailability of professional programmers with expertise in these languages. I am sure that by attracting some young minds to work in these programming languages, their eventual day of demise can be prolonged for a long time. END

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